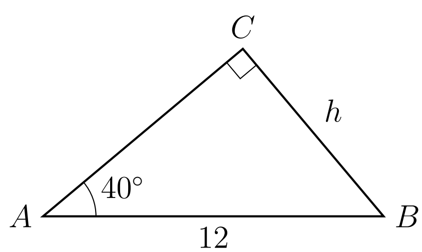
Handling the Ambiguous SSA Case Completely

Testing the height first, then reporting one triangle, two triangles, or none, and carrying both cases through applied problems

Directions. Every problem gives two sides and an angle that is *not* between them (SSA). Before solving, run the height test.

- Side a is opposite angle A, and b is the other given side. The perpendicular from C to the line through A and B has length $h = b \sin A$.
- $a < h$: no triangle. $a = h$: one right triangle. $h < a < b$: **two** triangles. $a \geq b$: one triangle.
- When two triangles exist, report *both* answers. The law of sines gives the acute angle; its obtuse partner is $180°$ minus it.
- Round lengths and angles to two decimal places, and state units.

1. A drawing calls for a triangle ABC with $b = AC = 12$ cm, $a = BC = 6$ cm and $A = 40°$. Drop the perpendicular from C to the line through A and B: in the right triangle it makes, the 12 cm side is the hypotenuse and the angle at A is $40°$.



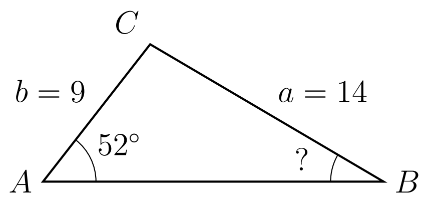
(a) Find the height h.

Answer: ________________ cm

(b) How many triangles fit the drawing? Justify your count by comparing h with the given sides.

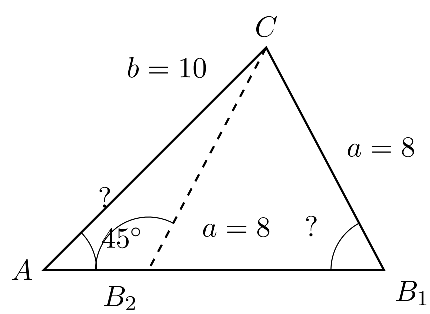
Answer: ________________

2. A triangle has $a = 14$ cm, $b = 9$ cm and $A = 52°$. Find angle B.



Answer: ________________ degrees

3. A triangle has $a = 8$ cm, $b = 10$ cm and $A = 45°$. Two triangles fit this data; the figure shows both.



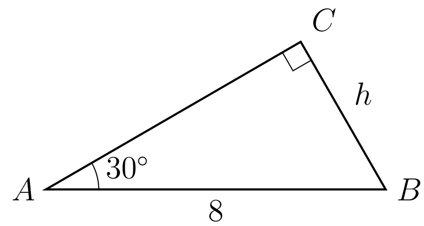
(a) Find the acute value of angle B.

Answer: ________________ degrees

(b) Find the obtuse value of angle B.

Answer: ________________ degrees

4. A triangle has $b = AC = 8$ cm, $a = BC = 5$ cm and $A = 30°$. The right triangle below has the 8 cm side as its hypotenuse and the angle at A equal to $30°$.



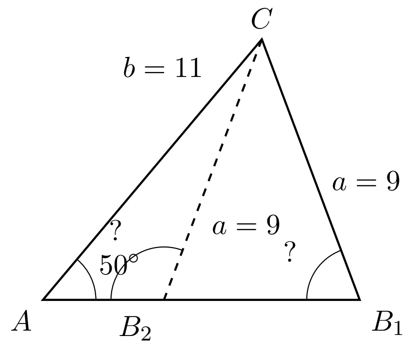
(a) Find the height h.

Answer: ________________ cm

(b) How many triangles fit this data? Justify your count.

Answer: ________________

5. A triangle has $a = 9$ cm, $b = 11$ cm and $A = 50°$.



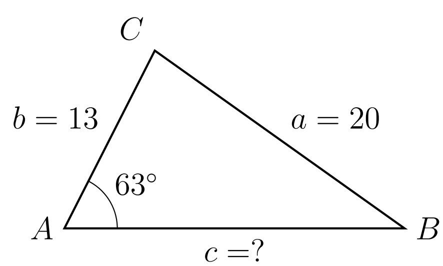
(a) Find the acute value of angle B.

Answer: ________________ degrees

(b) Find the obtuse value of angle B.

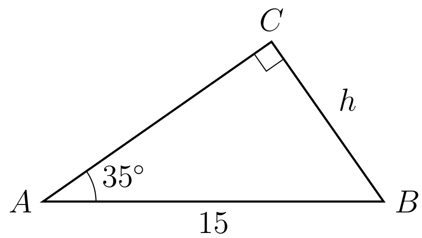
Answer: ________________ degrees

6. A triangle has $a = 20$ cm, $b = 13$ cm and $A = 63°$. Find the third side c.



Answer: ________________ cm

7. A triangle has $b = AC = 15$ cm, $a = BC = 7$ cm and $A = 35°$. The right triangle below has the 15 cm side as its hypotenuse and the angle at A equal to 35°.



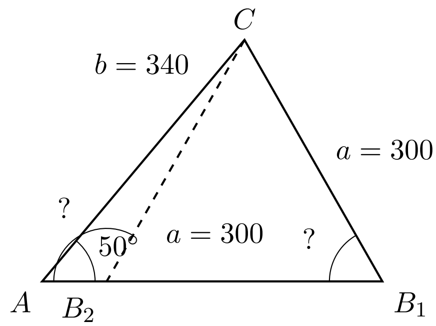
(a) Find the height h.

Answer: ________________ cm

(b) How many triangles fit this data? Justify your count.

Answer: ________________

8. A coastguard station at A tracks a launch at C, a distance $b = 340$ metres away. A buoy sits at B on a straight shoreline through A, and the launch is $a = 300$ metres from the buoy. The angle at the station, between the shoreline and the line to the launch, is $A = 50°$.



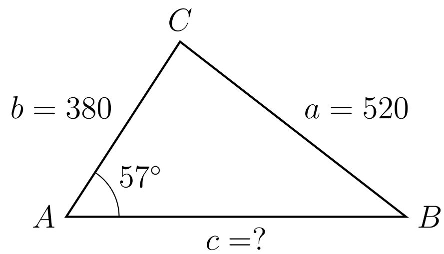
(a) Find the acute possible value of the angle B at the buoy.

Answer: ________________ degrees

(b) Find the obtuse possible value of the angle B.

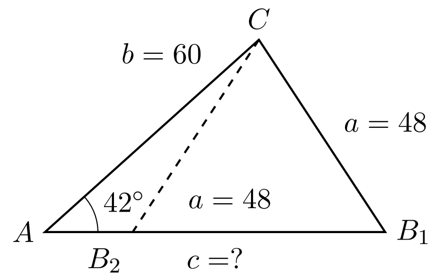
Answer: ________________ degrees

9. A surveyor at A measures $b = AC = 380$ metres to a rock at C, and finds the rock is $a = 520$ metres from a post at B on a straight fence line through A. The angle at A between the fence and the line to the rock is $57°$. Find c, the distance from A to the post, in metres.



Answer: ________________ m

10. A cable of length $b = 60$ metres runs from an anchor at A up to the top of a pole at C. A second cable of length $a = 48$ metres runs from C down to an anchor at B on the level ground line through A. The first cable makes an angle of $A = 42°$ with that ground line.



(a) Find the longer possible distance c from A to B, in metres.

Answer: ________________ m

(b) Find the shorter possible distance c, in metres.

Answer: ________________ m

(c) Explain what each answer describes physically, and why both are genuine solutions.

Answer: ________________